

ONE NODE - MINIMAX H3

Video + native stereo audio in one ComfyUI panel

USER GUIDE | VERSION 3.24

OPTIMIZED FOR NVIDIA RTX 16 GB

The protected 15-second recipes target 16 GB NVIDIA RTX cards using a 0.5 MP draft, model offload, FP32 latent upscaling, small temporal windows, CPU accumulation, and low-denoise refinement.

COMING SOON: 8 GB VRAM EDITION

A separate 8 GB optimized edition is in development. The current v3.24 release is not an 8 GB guarantee.

Prepared for the public GitHub release. Model weights are not included.

The New Game Plus | 18 September 2026

1. What this node does

One Node - MiniMax H3 writes and submits real ComfyUI graphs while keeping the model runtime in ComfyUI. It removes the need to wire a large graph by hand.

Mode	Use it for	Model family
T2V	Video and native audio from text.	FL2VA
I2V	Animate a first frame or bridge first and last frames.	FL2VA
R2V	Identity, style, motion, camera, and voice references.	Ref2VA
Studio	Compose references, continue shots, and assemble sequences.	Optional nodes
Finish / Repair	Recover detail or regenerate a masked region.	Optional nodes

H3 is guidance-distilled. Write the desired scene, action, camera, audio, and restrictions inside the main positive prompt.

Guide map

- 1 Install ComfyUI, this node, H3 models, and the protected-HD companions.
- 2 Select FL2VA or Ref2VA, then choose the matching mode.
- 3 Set aspect, duration, references, prompt, and quality recipe.
- 4 Approve a draft before spending time on a hero-quality render.

IMPORTANT

1080p and 2K keep the selected aspect ratio. A 9:16 job is vertical; a 16:9 job is horizontal.

2. Installation

Update ComfyUI first. It must provide native MiniMax H3 conditioning, video/audio nodes, samplers, masks, and AV latent utilities.

Install the wrapper

```
cd ComfyUI/custom_nodes
git clone https://github.com/Johnzx07/ComfyUI-MiniMaxH3-OneNode.git
```

Install the protected-HD companions

```
git clone https://github.com/ckinpx/ComfyUI-MMH3Tools.git
git clone https://github.com/LBH-123-AI/Comfyui_Minimax_h3_latent_upscaler.git
```

Run each companion package requirements file with the Python interpreter used by ComfyUI. Restart ComfyUI after installing or updating Python nodes.

Required H3 files

```
ComfyUI/models/
|-- diffusion_models/
|   |-- minimax_h3_fl2va_pruned_fp8_scaled.safetensors
|   |-- minimax_h3_ref2va_pruned_fp8_scaled.safetensors
|-- text_encoders/
|   |-- qwen3vl_32b_minimax_h3_int8_convrot.safetensors
`-- vae/
    |-- minimax_h3_video_vae_fp16.safetensors
    |-- minimax_h3_audio_vae_fp32.safetensors
```

Protected HD also needs `minimax_h3_latent_upscaler_3d_fp32.pth` in `ComfyUI/models/latent_upscale_models/`.

LICENSES

Weights are not included. Follow the MiniMax H3 Community License, the Comfy-Org model card, and each optional model or LoRA license.

3. Add the node and select models

- 1 Restart ComfyUI and hard-refresh the browser.
- 2 Search for One Node MiniMax H3 on the canvas.
- 3 Add One Node - MiniMax H3 (video + audio).
- 4 Open Models and select FL2VA or Ref2VA, the text encoder, and both VAEs.
- 5 Choose Rescan models after copying new files.

T2V PANEL MAP	CONTROL
Mode	T2V with FL2VA
Prompt	Integrated picture, camera, action, and sound
Video	Aspect ratio, megapixels, duration, and frame count
Sampling	Steps, sampler, scheduler, and seed
Advanced	Two-pass HD, LoRAs, acceleration, and memory controls
Models	Diffusion model, text encoder, video VAE, and audio VAE

The panel is resizable. Generate shows progress, step, and elapsed time. Finished clips appear in the Gallery.

4. T2V and I2V workflows

Text to video

- 1 Select T2V and the FL2VA diffusion model.
- 2 Set aspect ratio, duration, and draft resolution.
- 3 Describe the picture, camera, action, sound, dialogue, ambience, and music.
- 4 Generate a short draft before using the protected hero recipe.

Image to video and first-to-last

- 1 Select I2V and upload a first frame.
- 2 Optionally add a compatible last frame from the same shot.
- 3 Use subtle camera and subject motion when identity or line art must stay stable.
- 4 Use about 3-4 seconds for a tight first-to-last transition.

I2V PANEL MAP	CONTROL
First frame	Required visual anchor
Last frame	Optional ending state for the same shot
Reframe	Fit, crop, and output aspect handling
Motion	Describe small, deliberate camera and subject changes

PROMPT RULE

There is no separate negative prompt. State restrictions inside the main prompt, including exact subject and object counts.

5. Reference mode

Reference mode uses Ref2VA and can combine images, video, and audio. Use reference tags to define what each file contributes. Fewer strong references are easier to control.

Reference	Good use	Prompt example
Picture	Identity, object, costume, art style	Use for the same face and outfit.
Video	Camera path, motion rhythm, action	Match the timing of .
Audio	Voice timbre, dialogue, rhythm, ambience	Use the voice quality of .

- 1 Select R2V and the Ref2VA model.
- 2 Attach references and insert their exact tags.
- 3 Define subjects and reference roles before the shot.
- 4 Prefer five or fewer images unless more are essential.
- 5 Use max detail only when identity strength justifies the cost.

R2V REFERENCE MAP	CONTROL
Pictures	Identity, objects, wardrobe, and art style
Videos	Camera path, motion timing, and physical action
Audio	Voice timbre, dialogue guide, rhythm, and ambience
Reference detail	Match for speed; max for stronger identity

ANIME MOTION IN R2V

Anime Motion is beta in Reference mode. Begin at strength 0.65 and run a short identity and audio comparison before production.

6. Protected 15-second / 16 GB recipe

Open Advanced -> Two-pass HD and press 1080p or 2K. The button writes the complete known-good configuration.

Stage	Protected setting	Purpose
Draft	362 frames, 15 seconds, 0.5 MP	Establish motion within the VRAM target.
Latent upscale	FP32 model, temporal chunks, force unload	Increase latent size without frame decode.
AV protection	Split AV, lock audio, rejoin	Keep stage-1 audio out of denoise.
Refine	4 steps, 0.20 denoise, small windows	Recover detail with controlled change.
Decode	Single final decode	Create the final ratio-aware MP4.

Target sizes

- **1080p:** 2 MP, such as 1920 x 1088 or 1088 x 1920.
- **2K:** 4 MP, such as approximately 2688 x 1504.
- **Aspect:** select 9:16 first for vertical output.

PROTECTED PRESET SUMMARY	CONTROL
Draft	0.5 MP
Duration	15 seconds / 362 frames
Upscale	FP32 neural latent upscaler
Windows	Ultra Safe with CPU accumulation
Refine	4 steps at 0.20 denoise
Targets	Ratio-aware 1080p or 2K-class
HERO QUALITY	
Keep Turbo, PDD, and caches off unless the selected recipe explicitly requires them. Review every final clip for artifacts.	

7. LoRAs, speed, and quality

Turbo and style LoRAs use the companion H3 Turbo loader. Place files in ComfyUI/models/loras/ and rescan.

Option	Starting point	Best use
Anime Motion 1400	I2V strength 1.0	Anime motion with a supplied keyframe.
Anime Motion in R2V	Strength 0.65, beta	Reference-led animation after a short test.
LightX 4-step	Pinned 4-step recipe	Fast iteration and volume work.
LightX 8-step	Pinned 8-step recipe	More refinement at moderate cost.
Full 20-step	Euler + beta for hero work	Complex motion, identity, and finals.

Cache systems

TeaCache, Spectrum, FirstBlockCache, and EasyCache can accelerate suitable shots. Use one cache strategy at a time. Disable caches for fast motion, lip sync, and hero renders.

Attention and memory

Sage attention, Sol attention, feed-forward chunking, model offload, and CPU accumulation are separate controls. Verify one change at a time.

NOT TEXT TO IMAGE

Anime Motion is trained for H3 video behavior. It is not a text-to-image LoRA.

8. Prompt and audio checklist

Write one coherent timeline. Put picture and sound in the same prompt. For Reference mode, define subjects and every reference role first.

Recommended structure

```
integrated_multimodal_description:  
[Shot 1] Subject, composition, action, camera, light, and timing.  
  
overall_soundscape:  
Dialogue, room tone, ambience, sound effects, and synchronization.  
  
non_diegetic_music:  
Score, instrumentation, tempo, emotional arc, or N/A.
```

Before generating

- State exact subject count and stable identity details.
- Use a small number of clear actions.
- Name the camera move and speed.
- Describe ambience and effects even when there is no music.
- Request no visible text when accidental captions would be harmful.

After generating

- Watch the entire clip, including the first and last second.
- Check faces, hands, object count, line art, identity, continuity, and flicker.
- Listen for speech, ambience, sync, clipping, or a quiet track.
- Keep the seed and settings for successful shots.

9. Optional tools and credits

Optional tabs remain disabled when their required classes are missing. Install only the features you plan to use, then restart ComfyUI.

Feature	Upstream project
Protected HD windows and AV	ckinpx / ComfyUI-MMH3Tools
Latent upscale	LBH-123-AI / Minimax H3 Latent Upscaler
Turbo and LoRA loading	Larryvrh / ComfyUI-MiniMax-H3-Turbo
Studio and continuation	CGlide / ComfyUI-CGlide
Repair and context masks	ethanfel / MiniMaxH3 Contex Loop
Sage and Sol attention	kijai / KJNodes; Saganaki22 / sol-attn
Cache systems	Icyoung TeaCache; duckyshe11 FBC; xmarre Spectrum
Upscale tab	Comfy-Org RTX VSR; numz SeedVR2; naxci1 FlashVSR

FINISH AND UPSCALE MAP	CONTROL
H3 Finish	Source-locked latent detail recovery
RTX VSR	Fast NVIDIA hardware video super resolution
SeedVR2	Diffusion restoration with temporal stability
FlashVSR	Fast diffusion upscaling
ESRGAN	Standard per-frame upscale models

See CREDITS.md for exact links, node classes, model acknowledgements, and redistribution notes.

10. Troubleshooting and quick reference

Problem	Check
Node missing	Confirm custom_nodes placement, restart, and inspect the first console error.
Blank panel	Restart ComfyUI and hard-refresh after updating.
Models missing	Check folders and filenames, then rescan models.
Protected preset unavailable	Install MMH3Tools, the latent upscaler node, and its FP32 model.
CUDA out of memory	Use 0.5 MP draft, Ultra Safe windows, offload, and close other GPU apps.
Quiet or missing audio	Select the audio VAE, prompt sound, and review the exported MP4.
Identity drift	Use fewer references, lower motion, lower R2V Anime strength, and shorten the clip.

FAST START

I2V hero test: first frame + 5 seconds + full 20 steps. Approve motion first, then use protected 1080p or 2K for the final.

Output folder

ComfyUI/output/ComfyUI-MiniMaxH3-OneNode/

Release status

- Optimized for NVIDIA RTX 16 GB cards.
- Protected 15-second 1080p and 2K-class presets included.
- 8 GB optimized edition: coming soon.
- No model weights, user files, tokens, or personal render history are bundled.

MiniMax H3 and companion projects retain their own licenses. This wrapper is Apache-2.0. See LICENSE, NOTICE, CREDITS.md, and INSTALL.md.